



Incarceration as a unique social stressor during pregnancy: Implications for maternal and newborn health

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ABSTRACT

Rationale: Exposure to incarceration during pregnancy is a stressful life event that poses risk for both maternal and newborn health.

Objective: We conducted a pooled, cross-sectional analysis of the link between incarceration exposure of mothers during pregnancy – personally or vicariously through a romantic partner – and maternal and newborn health using data from the Pregnancy Risk Assessment Monitoring System from years 2012–2015.

Method: We applied a novel approach that compares mothers who were exposed to incarceration to the strategic comparison group of mothers who were not exposed to incarceration, but experienced varying numbers of stressors.

Results: The initial findings indicated that respondents exposed to incarceration exhibited reductions in maternal and newborn health relative to respondents not exposed to incarceration. However, these differences were substantially attenuated when comparing the incarceration-exposed group to a more comparable stressor-exposed reference group.

Conclusions: While incarceration exposure has adverse consequences for health outcomes of mothers and newborns, incarceration exposure may not carry unique implications for maternal and newborn wellbeing beyond the clustering of other stressful life events. Policies targeting incarceration exposure as a means of promoting health must also provide adequate attention to co-occurring stressors.

1. Introduction

Incarceration is widely recognized as a critical public health concern (Dumont et al., 2012; Wildeman and Wang, 2017). Indeed, a growing body of research has repeatedly linked incarceration to worsening health, both for individuals (Blankenship et al., 2018; Massoglia and Pridemore, 2015; Massoglia and Remster, 2019) and family members such as spouses and children (Lee et al., 2014; Wildeman et al., 2018). The health consequences of incarceration exposure during pregnancy for mothers and newborns, however, are less often examined. Even so, the findings of this small but growing body of literature reveal lower quality prenatal care (Dumont et al., 2014), higher prenatal cigarette consumption (Dumont et al., 2015), and a greater risk of food insecurity (Testa and Jackson, 2019) among expectant mothers exposed to incarceration during pregnancy. Their infant children, moreover, appear to be at risk of infant mortality (Wildeman, 2012), although

some research finds no link between incarceration exposure during pregnancy and select newborn health outcomes, such as low birth weight (Dallaire et al., 2018).

While there are several perspectives that elucidate the association between incarceration exposure and adverse health outcomes among individuals and their families, the leading paradigm is the stress process perspective (Massoglia, 2008; Pearlin, 1989; Pearlin et al., 1997; Thoits, 2010). The stress process perspective proposes that individuals from disadvantageous contexts are more likely to experience stressors (including incarceration) than those from advantageous contexts. Such stressors, moreover, are expected to have significant consequences for physical and mental health (Pearlin, 1989; Pearlin et al., 1997; Thoits, 2010). Furthermore, it is argued that stress can proliferate in two ways – within individuals (e.g., within-individual stress proliferation) and between individuals (e.g., stress contagion) – and can therefore impact the health of not only those who directly experience the stressor, but

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also those intimately connected to persons directly experiencing the stressor (e.g., spouse, children, etc.). While a person is incarcerated, within-individual stress proliferation can occur in the form of deprivations of liberty and autonomy, a lack of privacy, exposure to violence, and hostile interactions with correctional staff (Goffman, 1963; Porter, 2019; Turney, 2014). Similarly, stress contagion likely plays a role in the health of spouses and children, as incarceration typically results in limited communication and persistent worrying about the safety of one's partner/parent (Comfort, 2008; Lee et al., 2014). Following release from incarceration, both within-individual stress proliferation and stress contagion can stem from various legal restrictions on employment and housing, the social stigma of incarceration, and challenges with reestablishing personal relationships (Rice et al., 2018; Travis et al., 2014; Western et al., 2015).

Although the stress process perspective serves as a framework for understanding incarceration as a unique and consequential stressor with important health implications (Massoglia and Pridemore, 2015; Pearlin, 1989; Thoits, 2010; Turney, 2014), the applicability of the stress process perspective is in no way restricted to incarceration. This is certainly the case when examining the health of expectant mothers during pregnancy and their newborn children. For instance, stressful life events (e.g., experiencing financial hardship, abuse by a partner, family conflict, marital/relationship discord) are linked to adverse prenatal and maternal health outcomes, such as lower quality prenatal care (Kitsantas et al., 2012), gestational diabetes (Hosler et al., 2011), and post-partum depression (Stone et al., 2015). A similar set of stressors appear to elevate the risk of several infant health outcomes including preterm birth (Dole et al., 2003; McLemore et al., 2018), low birth weight (Borders et al., 2007), higher cortisol levels (Romero-Gonzalez et al., 2018), greater number of infant illnesses and hospital visits (Phelan et al., 2015), and inhibited infant development (Entringer et al., 2015). Thus, work that aims to classify incarceration as a potent stressor linked to adverse health outcomes for mothers and newborns must simultaneously consider a host of stressors that are correlated with, but not inextricably tied to incarceration.

Prior research on whether experiencing incarceration carries unique harms typically draws on two competing perspectives: “unique effects” and “selection effects” (Shlafer, Poehlmann and Donelan-McCall, 2012). The unique effects perspective posits that incarceration directly causes negative outcomes. Consequently, incarceration should yield an independent association with health outcomes of interest, even after adjusting for other stressful life events. Indeed, the extant literature on the consequences of incarceration often conceptualizes incarceration as a unique and potent stressor (Travis et al., 2014) “that requires dramatic changes in a short period across a number of life circumstances” (Massoglia and Pridemore, 2015: p. 299). For example, Dallaire et al. (2015: p. 110) study of maternal incarceration notes that “as a risk mechanism, maternal incarceration confers *unique* or specific risk” [emphasis added]. Previous studies assessing the consequences of incarceration during pregnancy typically reveal that incarceration exposure is linked to adverse health outcomes among mothers and their infants, and that this relationship is only partially attenuated after accounting for other stressors (Dallaire et al., 2018; Dumont et al., 2014, 2015; Testa and Jackson, 2019; Wildeman, 2012).

Alternatively, the “section effects” perspective proposes that those who experience incarceration are fundamentally different from those who do not, and risk factors associated with incarceration – not incarceration itself – are responsible for any observed adverse outcomes (Shlafer et al., 2012). For instance, Shlafer et al. (2012) find that significant associations between maternal criminal justice contact and adolescent health risk behaviors became null after adjusting for prenatal risk factors. However, adequately testing this perspective requires researcher to adjust for a host of co-occurring risk factors, many of which are often unavailable in survey data (Porter and King, 2015). Thus, while the approaches in past research are informative, most research aiming to differentiate between “unique” and “selection” effects

related to incarceration do so by adjusting statistical models for control variables. We contend that prior work has been methodologically limited in this regard, as isolating the health impact of incarceration from that of other stressful life events has proven challenging. This problem is probably especially acute when using standard regression techniques that are meant to correct for small differences between groups (Murnane and Willett, 2010), particularly in light of the sizeable differences between incarcerated and non-incarcerated populations across a number of observable and unobservable characteristics (Travis et al., 2014). As Wildeman and Muller (2012, p. 14) have previously pointed out “concerns about selection bias suffuse social science, but they are particularly acute in this area, where there are strong theoretical and empirical reasons to believe that any effect of incarceration is solely attributable to negative endowments that increased individual's likelihood of incarceration in the first place.” An alternative approach is to facilitate strategic comparisons between persons who are similarly exposed to stressors, but differentially exposed to incarceration (Apel and Sweeten, 2010; Massoglia et al., 2011; Porter, 2014; Porter and King, 2015). Such strategic comparisons are crucial, as they can elucidate whether incarceration as a social stressor has unique health consequences for mothers and their newborns above and beyond the health consequences of stressors commonly, but not inevitably, linked to incarceration (e.g., abuse, unemployment, family conflict, etc.) (Mersky et al., 2018).

To address this issue, the current study proposes a novel approach, using a stressor-informed strategic comparison group design that compares individuals exposed to incarceration to those who were not exposed to incarceration, but experienced differential exposure to financial, spousal, or emotional stressors. This approach helps account for unobservable characteristics that may be associated with both experiencing stressful life events and adverse health outcomes, while also providing a more precise assessment of whether incarceration has unique health consequences among mothers and their newborns beyond those linked to other stressful life events.

2. Method

2.1. Data

Data for the current study are drawn from the Pregnancy Risk Assessment Monitoring System (PRAMS). PRAMS is a state-based survey administered annually by the Centers for Disease Control and Prevention in collaboration with state health departments. Participating states conduct a stratified random sample of approximately 1000 to 3000 women who are residents of the participating state and have delivered a live birth within the state. Retrospective data are gathered via surveys that are mailed to mothers' homes 2–6 months following the live birth. States make up to five contact attempts via mail and follow up with non-responders by phone call within a week of the last mailed survey. To maximize response rates, up to 15 calling attempts are made over different times of day and varying days of the week (Shulman et al., 2018).

The PRAMS survey asks a variety of questions regarding maternal attitudes, physical and mental wellbeing, and health behaviors before, during, and after birth. Additionally, responses of mothers who complete the questionnaire are linked to selected birth certificate data. At present, the PRAMS survey covers approximately 83% of all births in the United States and is the only surveillance system that provides data about pregnancy and the first few months after birth (Shulman et al., 2018). PRAMS data were approved for use in this study by the Centers for Disease Control and Prevention.

2.2. Dependent variables

2.2.1. Prenatal maternal health

Kessner index classifies the adequacy of prenatal care based on the

information derived from birth certificates, including the month of pregnancy in which prenatal care began, the number of prenatal care visits, and the length of pregnancy (Allen et al., 2019; Kessner et al., 1973; Danilack et al., 2019; Huber et al., 2018). We examined categories within the Kessner index to assess variation in prenatal care, including adequate, intermediate, and inadequate. Classifications of the adequacy of care based on the Kessner index are provided in [supplemental Appendix A](#). Maternal smoking is measured using self-reported information on the total number of cigarettes smoked per day across each trimester of pregnancy. Responses regarding cigarette use during each trimester were averaged and cigarette consumption was classified into three distinct categories: none (0 cigarettes daily), low (10 or less cigarettes daily), and high (more than 10 cigarettes daily) (Dumont et al., 2015).

2.2.2. Postnatal maternal health

Post-partum depression symptoms are a binary item coded as “yes” if respondents answered “always” or “often” to either of the following questions: (1) “since your new baby was born how often have you felt depressed or hopeless?” And (2), “since your new baby was born, how often have you had little interest or little pleasure in doing things?” Post-partum depression refers to “a non-psychotic depressive episode that begins in or extends into the postpartum period” (O’Hara and Swain, 1996: p. 38). Post-partum depression is multifaceted disorder with symptoms including dramatic changes in mood, irritability, hypersensitivity, insomnia, anxiety, depressed mood, and psychotic thoughts (O’Hara and Segre, 2008; O’Hara, 2009). While the classification employed in this study is consistent with the procedure in prior PRAMS research (Hayes et al., 2010; Ko et al., 2017), given the complexity of postpartum depression, this measure may not be inclusive of all aspects of the disorder.

2.2.3. Newborn health

Low birth weight (LBW) measures whether a newborn’s birth weight was less than 2500 g (1 = Yes; 0 = No). *Small for gestational age (SGA)* is a binary variable that identifies newborns who were below the 10th percentile for their gestational age (1 = Yes; 0 = No). *Preterm birth* is a binary variable representing births before 37 weeks (1 = Yes; 0 = No). *Neonatal intensive care unit (NICU)* measures whether a newborn spent time in the intensive care unit following birth (1 = Yes; 0 = No). *Extended hospitalization* is a binary variable indicating whether the length of a newborn’s hospital stay was 6 days or more following birth (1 = Yes; 0 = No). *Newborn Health Index* sums each of the above items into a single index ranging from 0 to 5 (Cronbach’s $\alpha = 0.799$).

2.3. Independent variables

The focal independent variables measure a variety of distinct stressful life events that occurred in the 12 months prior to birth, and therefore captures events occurring slightly before or during pregnancy. *Incarceration exposure* is measured using a survey item that asks respondents whether in the 12 months before to birth, “I or my husband or partner went to jail” (1 = Yes; 0 = No). Although the question inquiries about jail, the measure is used as a proxy for incarceration in any correctional facility as the terms jail and prison are used interchangeably among the general public (Dumont et al., 2015). Further, although this item measures recent mothers who were exposed to incarceration during their pregnancy either personally or vicariously through their husband/partner we hereafter refer to this population as “incarcerated” for simplicity.

Next, we generate separate stressor categories based on responses to 11 binary items in which respondents indicated whether they had experienced a specific stressor in the 12 months prior to birth. The Cronbach’s α for the 11 life events was 0.66, indicating fairly good reliability of the items. We then classified stressors into distinct thematic groupings based on classifications used in previous PRAMS

research (Lu and Chen, 2004; Nkansah-Amankra et al., 2010): *Financial stress* (respondents husband/partner lost their job, respondent lost their job, respondent had bills that couldn’t be paid, respondent was homeless), *spousal stress* (respondent got separated/divorced from husband/partner, respondent argued a lot with husband/partner, husband/partner did not want pregnancy, respondent was physically abused by husband/partner during pregnancy), and *emotional stress* (a close family member was very sick, someone very close had died, someone very close had a problem with drugs or alcohol). To examine heterogeneity in stressor exposure within each stressor group, we created distinct categories of no, low, medium, and high stressor exposure. [Supplemental Appendix B](#) details the criteria for each stressor category.

2.4. Control variables

We control for several characteristics of the mother including maternal age at birth (≤ 17 [reference category], 18–24, 25–29, 30–23, 35+), maternal race (white [reference category], black, Hispanic, other race), college graduate (1 = Yes, 0 = No), currently married (1 = Yes, 0 = No), the number of previous births (0 [reference category], 1, 2, 3+), maternal weight gain during pregnancy (pounds), whether a mother reported that she was planning to get pregnant (1 = Yes, 0 = No), and categories of pre-pregnancy body mass index (BMI) (underweight [reference category], normal weight, overweight, obese). Finally, we include a measure of stressors faced by a mother in the 12 months prior to birth based on the stressor items discussed above.

2.5. Analytic plan

First, we employ a more traditional methodological approach by comparing individuals who had incarceration exposure to those who did not, while controlling for confounding variables including stressors. Subsequently, we apply the novel approach of comparing incarcerated mothers to our strategic comparison group of persons who were not incarcerated but had varied exposure to diverse financial, spousal, or emotional stressors (i.e. none, low, medium, high), while controlling for confounding variables and other stressor categories. This approach enables a more fine-tuned investigation of whether incarceration serves as a particularly potent stressor with unique effects on maternal and newborn health.

To assess the binary outcome variables (post-partum depression, low birth weight, small gestational age, NICU, preterm birth, and extended hospitalization), we employ logistic regression. For the newborn health index, we use negative binomial regression, given that the variable is positively skewed and over-dispersed. For the categorical outcomes (Kessner index and maternal smoking) we use multinomial logistic regression. All statistical analyses are conducted using Stata V.15.1. Models were adjusted for survey weights and strata information. Approximately 17.4% of the sample had missing data on at least one variable. Missing data were addressed using multiple imputation with chained equations, resulting in the utilization of 10 multiply imputed data sets in the current analysis.

3. Results

[Table 1](#) presents the weighted descriptive statistics for the analytic sample. Approximately 3.8% of mothers reported incarceration in the 12 months prior to birth. Compared to respondents without incarceration, the incarceration sample was more likely to experience inadequate prenatal care (10.5% vs. 5.5%), smoke more than 10 cigarettes per day on average during pregnancy (13.6% vs. 3.3%), and report post-partum depression symptoms (25.6% vs. 11.6%). Across newborn health outcomes, the incarceration group were more likely to report low birthweight (10.2% vs. 6.9%), small for gestational age (13.4% vs. 9.9%), preterm birth (11.0% vs. 8.3%), time spent in NICU (16.4% vs. 12.0%), extended newborn hospitalization (11.8% vs.

Table 1
Summary statistics by category of stressor.

Variables		Incarceration Exposure		Financial Stressors			Spousal Stressors			Emotional Stressors					
		Yes	No	None	Low	Medium	High	None	Low	Medium	High	None	Low	Medium	High
Kessner Index															
Adequate		51.8%	67.5%	69.9%	63.6%	57.8%	51.4%	69.8%	62.1%	58.1%	54.6%	67.7%	67.4%	67.6%	62.5%
Intermediate		27.9%	19.2%	17.8%	21.6%	25.2%	29.3%	17.9%	22.6%	24.9%	27.2%	19.2%	19.1%	18.7%	24.3%
Inadequate		10.5%	5.5%	4.8%	6.4%	8.7%	11.0%	4.9%	7.0%	7.4%	9.9%	5.3%	5.9%	5.6%	5.2%
Maternal Smoking															
None		70.0%	91.7%	94.1%	87.4%	82.1%	76.3%	93.6%	87.6%	82.1%	78.5%	93.8%	89.5%	86.4%	75.7%
Low		16.4%	5.0%	3.7%	7.4%	10.0%	13.4%	3.9%	7.6%	10.9%	12.3%	3.9%	6.2%	7.8%	14.5%
High		13.6%	3.3%	2.2%	5.2%	7.9%	10.4%	2.5%	4.8%	7.1%	9.2%	2.3%	4.3%	5.8%	9.7%
Maternal Depression															
Post-Partum	Depression	25.6%	11.6%	9.1%	16.3%	20.9%	27.6%	8.5%	19.0%	25.0%	29.9%	10.0%	13.4%	15.6%	25.0%
Newborn Health															
LBW		10.2%	6.9%	6.6%	7.6%	7.9%	9.9%	6.6%	7.3%	9.3%	10.2%	6.8%	7.0%	7.5%	7.7%
SGA		13.4%	9.9%	9.4%	10.6%	11.8%	13.8%	9.5%	10.7%	11.3%	14.8%	9.6%	10.0%	11.2%	12.9%
Preterm		11.0%	8.3%	7.9%	9.2%	9.4%	9.9%	8.0%	9.0%	9.6%	9.6%	8.2%	8.6%	8.4%	8.4%
NICU		16.4%	12.0%	11.3%	13.7%	14.3%	16.2%	11.5%	13.1%	14.2%	17.0%	11.6%	12.4%	13.1%	16.9%
Hospitalization		11.8%	7.2%	6.6%	8.6%	8.6%	11.3%	6.8%	7.8%	9.3%	10.6%	6.8%	7.8%	8.0%	9.9%
Health Index		.628	.443	.418	.497	.519	.611	.424	.479	.537	.622	.429	.458	.481	.558

Note. Columns may not sum to 100% due to rounding.

Abbreviations: LBW = Low birth Weight; SGA = Small for gestational age; NICU = Neonatal intensive care unit.

7.2%), and worse overall newborn health (0.628 vs. 0.442). [Supplemental Appendix C](#) provides the summary statistics of the control variables for the analytic sample stratified by incarceration.

In addition to the disparities in health outcomes between those differentially exposed to incarceration, the results in [Supplemental Appendix D](#) reveal that the incarcerated sample was more than 14 times as likely to experience a high number of stressors compared to non-incarcerated participants (e.g., 0.234 vs. 0.016), and approximately 10 times as likely to experience a high number of financial (e.g., 0.170 vs. 0.019) and spousal stressors (e.g., 0.179 vs. 0.018). These results suggest that there are sizable differences in the extent to which incarcerated and non-incarcerated individuals are exposed to stressors, buttressing the need for health-focused analyses that employ more strategic comparisons between those incarcerated and those exposed to stressors in the absence of incarceration.

[Table 2](#) presents the results of the series of regression models assessing incarceration on prenatal healthcare, maternal smoking, and post-partum depression. The top row presents the results using the strategy commonly employed in prior research, in which incarcerated respondents are compared to those who did not report incarceration, while adjusting for relevant control variables and other stressful life events. The results of the multinomial logistic regression models indicate that incarceration is associated with a higher likelihood of intermediate ($RRR = 1.17$, 95% $CI = 1.03, 1.33$) or inadequate prenatal care ($RRR = 1.29$, 95% $CI = 1.07, 1.56$) relative to receiving adequate prenatal care. Additionally, incarcerated mothers are more likely to have low ($RRR = 1.59$, 95% $CI = 1.22, 2.09$) and high ($RRR = 2.03$, 95% $CI = 1.50, 2.77$) daily cigarette smoking relative to no smoking during pregnancy.

Subsequently, we adjust the reference category to respondents who were not incarcerated but nevertheless experienced varying numbers of financial, spousal, or emotional stressors. When comparing the incarcerated group to the non-incarcerated group, 4 out of 5 coefficients (80%) resulted in a positive association; however, when changing the reference group to those who experienced stressors in the absence of incarceration, the impact of incarceration is substantially attenuated. For instance, when examining comparisons between incarceration and the high financial stress reference category, 5 out of 5 incarceration coefficients yielded a null association. Compared to the high spousal stress category, 4 out of 5 coefficients have a null association (80%), with 1 negative association (20%). Finally, in comparison to the high emotional stressor category 2 out of 5 coefficients have a null

association (40%), 2 coefficients have a positive association (40%), and 1 coefficient yields a negative association (see [Tables 2 and 4](#)).

[Table 3](#) presents the results of the association between incarceration exposure and newborn health. The results demonstrate that, incarceration has a positive and statistically significant association with adverse newborn health outcomes in 5 out of 6 models (83.3%), including low birth weight ($OR = 1.15$, 95% $CI = 1.02, 1.30$), preterm birth ($OR = 1.16$, 95% $CI = 1.00, 1.34$), NICU ($OR = 1.14$, 95% $CI = 1.00, 1.30$), extended hospitalization ($OR = 1.30$, 95% $CI = 1.13, 1.51$), and worse overall newborn health ($IRR = 1.16$, 95% $CI = 1.07, 1.25$). After changing the reference category to individuals who experienced varying numbers of financial, spousal, or emotional stressors, the unique adverse effects of incarceration on newborn health begin to diminish. [Table 4](#) shows that when compared to respondents who experience a high number of stressors, the association between incarceration and newborn health is null across all outcomes.

In sum, across all maternal and infant health outcomes, when employing a more rigorous and comparable reference group, the majority of the coefficients are null, with significant coefficients showing no consistent trend in the commonly hypothesized, positive direction. [Supplemental Appendices E and F](#) report the results when comparing incarceration to specific stressful life events. The adverse maternal and newborn health consequences of incarceration are most commonly attenuated when compared to divorce, homelessness, and physical abuse.

4. Discussion

The present analysis takes steps to expand upon previous work on the adverse consequences of incarceration for health of mothers and newborns by using stressor-informed strategic comparison groups ([Apel and Sweeten, 2010](#); [Porter, 2014](#); [Porter and King, 2015](#)). Overall, we found that even though incarceration exposure during pregnancy was initially associated with a host of negative maternal and newborn health outcomes, these differences were attenuated (often to statistical non-significance) when a high-stress comparison group was employed. Thus, while initial results were consistent with stress proliferation theory as applied to newborn and maternal health, findings also suggest that, as a stressor, incarceration may not carry unique implications for maternal and newborn wellbeing when compared to individuals who were not previously incarcerated, but experienced high levels of stressors.

On one hand, the results suggest that incarcerated mothers as a

Table 2
Association between incarceration exposure and maternal health ($N = 117,240$).

Variable and comparison	Kessner Index		Maternal Smoking		Depression
	Intermediate vs. Adequate RRR (95% CI)	Inadequate vs. Adequate	Low vs. None RRR (95% CI)	High vs. None	Post-Partum Depression OR (95% CI)
Incarceration Exposure					
Inc vs. No Inc	1.17* (1.03, 1.33)	1.29** (1.07, 1.56)	1.59** (1.22, 2.09)	2.03*** (1.50, 2.77)	1.04 (.92, 1.19)
Financial Stress					
Inc vs. No Financial Stress	1.36*** (1.19, 1.55)	1.63** (1.34, 1.98)	1.88*** (1.41, 2.50)	2.75*** (1.99, 3.81)	1.02 (.81, 1.29)
Inc vs. Low Financial Stress	1.24** (1.09, 1.42)	1.48*** (1.20, 1.81)	1.76*** (1.31, 2.35)	2.22*** (1.63, 3.12)	1.06 (.85, 1.32)
Inc vs. Medium Financial Stress	1.06 (.91, 1.23)	1.09 (.87, 1.37)	1.55** (1.12, 2.15)	1.80** (1.24, 2.62)	1.18 (.92, 1.52)
Inc vs. High Financial Stress	.89 (.73, 1.08)	.87 (.66, 1.15)	.97 (.64, 1.46)	.98 (.61, 1.58)	.92 (.64, 1.32)
Spousal Stress					
Inc vs. No Spousal Stress	1.28*** (1.12, 1.46)	1.40** (1.15, 1.69)	1.73*** (1.31, 2.29)	2.15*** (1.57, 2.95)	1.82*** (1.60, 2.07)
Inc vs. Low Spousal Stress	1.13 (.99, 1.30)	1.22 (.99, 1.50)	1.55** (1.16, 2.08)	1.96*** (1.41, 2.73)	.99 (.86, 1.13)
Inc vs. Medium Spousal Stress	1.16† (.99, 1.36)	1.40** (1.09, 1.78)	1.63** (1.15, 2.32)	2.18*** (1.45, 3.29)	.84* (.72, .98)
Inc vs. High Spousal Stress	1.09 (.89, 1.35)	1.12 (.83, 1.51)	1.28 (.82, 1.99)	1.62 (.96, 2.74)	.76** (.62, .93)
Emotional Stress					
Inc vs. No Emotional Stress	1.13 (.99, 1.28)	1.26* (1.04, 1.52)	1.81*** (1.37, 2.38)	2.40*** (1.74, 3.31)	1.21** (1.06, 1.38)
Inc vs. Low Emotional Stress	1.22** (1.06, 1.39)	1.25* (1.02, 1.53)	1.66** (1.24, 2.22)	2.09*** (1.50, 2.93)	1.10 (.96, 1.26)
Inc vs. Medium Emotional Stress	1.25** (1.08, 1.44)	1.31* (1.04, 1.64)	1.43* (1.05, 1.95)	1.70** (1.19, 2.44)	.98 (.84, 1.13)
Inc vs. High Emotional Stress	1.06 (.87, 1.31)	1.63** (1.17, 2.26)	1.22 (.80, 1.87)	1.64* (1.03, 2.60)	.79* (.64, .97)

Note. All models include controls for maternal age at birth, maternal race, whether mother graduated college, maternal marital status, number of prior births, weight gain during pregnancy, mother planned pregnancy, pre-pregnancy BMI, and co-occurring stressors. Smoking model controls for smoking prior to pregnancy. Depression model controls for depression before pregnancy. Models uses data from eligible participating states from 2012 to 2015. * $p < .05$; ** $p < .01$; *** $p < .001$.

Abbreviations: RRR = Relative risk ratio; OR = Odds Ratio; Inc = Incarceration.

population are vulnerable to several adverse maternal and newborn health outcomes. Indeed, when compared to non-incarcerated mothers who experienced low or moderate levels of stressors, incarceration often still yields an independent association with adverse health outcomes. This pattern suggests that incarceration is a harmful stressor for maternal and newborn health. Still, incarceration rarely yielded an independent association on the outcomes of interest when compared to the high stressor reference category. This result suggests that, with few exceptions, addressing incarceration exposure as a means of promoting health among expectant mothers and newborns may have limited efficacy if co-occurring stressors – particularly physical abuse, divorce, and lack of adequate housing – are not given adequate attention. Considering that, in many cases, incarcerated participants exhibited a similar increase in the likelihood of most adverse maternal and newborn health outcomes as individuals exposed to high levels intersecting stressors in the absence of incarceration is telling. Ultimately, our key findings suggest initiatives focused exclusively on minimizing exposure to incarceration – as a more narrow, targeted endeavor – may be unlikely to serve as a panacea in the effort to eliminate health inequities. Instead, recognizing that incarcerated populations often experience a variety of stressful life-experiences, initiatives aimed at promoting health equity among the disadvantaged and disenfranchised must be broad in scope and take meaningful steps to address stressors in multiple facets of family life, including financial and spousal stressors, in addition to the hardships that stem from a period of incarceration.

While the present study is limited in terms of making causal inferences, there remains several important implications that flow from our analysis. Perhaps foremost, narrow criminal justice policies that focus attention on punishment of perceived dangerous classes and disregard health and well-being of families may be inefficient

approaches. Rather, health policies that are closely tied with social inequities are more likely to positively impact women of childbearing age who are most likely to encounter the justice system, either personally or through a partner. Moreover, since high-risk health conditions experienced during pregnancy have been linked to problem behaviors among offspring (Arseneault et al., 2002; Jackson and Vaughn, 2018), the focus on reducing stress proliferation among these women can also potentially decrease future offspring contact with formal systems of justice. Because criminal justice institutions are embedded in and intimately connected to larger currents of policy, sound health and social policy is also useful criminal justice policy. In addition to these conceptual implications, practical strategies that are empirically tested such as the Nurse Family Partnership that provides home visiting for new parents, parenting programs, and expanding early childhood education are potentially useful (Chase-Lansdale and Brooks-Gunn, 2014; Shonkoff and Fisher, 2013), especially given the high costs of incarceration and its collateral consequences. In addition, diversionary practices and transitional care planning programs can be implemented, as has been done with high health risk individuals (Teixeira et al., 2015). Our findings also have implications for future investigations of stress proliferation among incarcerated mothers. Specifically, life-course designs are needed that follow at-risk mothers during sensitive periods prior to and following incarceration with respect to health assessments and stressful life events.

4.1. Limitations

We highlight three primary limitations with the current study upon which future research can build. First, the incarceration item in the PRAMS survey does not specify whether incarceration was experienced

Table 3
Association between incarceration exposure and newborn health ($N = 117,240$).

Variable and comparison	LBW	SGA	Preterm	NICU	Hospital	Newborn Health Index
	OR (95% CI)	OR (95% CI)	OR (95% CI)	OR (95% CI)	OR (95% CI)	IRR (95% CI)
Inc vs. No Inc	1.15* (1.02, 1.30)	1.09(.94, 1.26)	1.16* (1.00, 1.34)	1.14* (1.00, 1.30)	1.30*** (1.13, 1.51)	1.16*** (1.07, 1.25)
Financial Stress						
Inc vs. No Financial Stress	1.21** (1.07, 1.37)	1.19* (1.03, 1.40)	1.22** (1.06, 1.42)	1.27** (1.11, 1.45)	1.47*** (1.27, 1.71)	1.25*** (1.15, 1.35)
Inc vs. Low Financial Stress	1.17* (1.03, 1.33)	1.12 (.96, 1.30)	1.14 (.98, 1.33)	1.10 (.96, 1.26)	1.25** (1.07, 1.46)	1.14** (1.05, 1.23)
Inc vs. Medium Financial Stress	1.15* (1.00, 1.32)	1.01 (.84, 1.19)	1.14 (.97, 1.35)	1.07 (.92, 1.24)	1.29** (1.09, 1.53)	1.12* (1.02, 1.22)
Inc vs. High Financial Stress	.96 (.79, 1.16)	.85 (.68, 1.06)	1.11 (.89, 1.39)	.96 (.78, 1.18)	.99 (.79, 1.23)	.98 (.87, 1.10)
Spousal Stress						
Inc vs. No Spousal Stress	1.20** (1.06, 1.36)	1.14(.98, 1.32)	1.18 (1.02, 1.37)	1.22** (1.06, 1.39)	1.36*** (1.17, 1.57)	1.21***(.12, 1.31)
Inc vs. Low Spousal Stress	1.23** (1.09, 1.40)	1.10 (.95, 1.29)	1.14 (.98, 1.33)	1.18* (1.02, 1.35)	1.36*** (1.17, 1.59)	1.19***(.109, 1.29)
Inc vs. Medium Spousal Stress	1.08 (.93, 1.25)	1.15 (.96, 1.39)	1.15 (.97, 1.37)	1.15 (.98, 1.36)	1.25* (1.04, 1.49)	1.14** (1.04, 1.26)
Inc vs. High Spousal Stress	1.00 (.83, 1.20)	.87 (.86, 1.10)	1.19 (.95, 1.48)	.96(.78, 1.19)	1.13 (.88, 1.45)	1.02 (.89, 1.15)
Emotional Stress						
Inc vs. No Emotional Stress	1.17* (1.03, 1.32)	1.15 (.99, 1.33)	1.16* (1.00, 1.35)	1.22** (1.07, 1.40)	1.41*** (1.22, 1.64)	1.21*** (1.11, 1.30)
Inc vs. Low Emotional Stress	1.17* (1.03, 1.33)	1.13 (.97, 1.32)	1.13 (.97, 1.32)	1.18* (1.03, 1.36)	1.30** (1.11, 1.51)	1.17*** (1.08, 1.27)
Inc vs. Medium Emotional Stress	1.10 (.96, 1.26)	1.01 (.85, 1.19)	1.18* (1.00, 1.39)	1.12 (.97, 1.30)	1.27** (1.08, 1.51)	1.12** (1.03, 1.23)
Inc vs. High Emotional Stress	1.16 (.96, 1.40)	.95 (.74, 1.21)	1.24 (.97, 1.59)	.88 (.71, 1.10)	1.10 (.86, 1.41)	1.05 (.93, 1.19)

Note. All models include controls for maternal age at birth, maternal race, whether mother graduated college, maternal marital status, number of prior births, weight gain during pregnancy, mother planned pregnancy, pre-pregnancy BMI, and co-occurring stressors. Models uses data from eligible participating states from 2012 to 2015. * $p < .05$; ** $p < .01$; *** $p < .001$.

Abbreviations: OR = Odds Ratio, IRR = incidence rate ratio; Inc = Incarceration; LBW = Low birth Weight; SGA = Small for gestational age; NICU = Neonatal intensive care unit.

by the respondent or her husband/partner. Since the majority of incarcerated persons in the United States are male, it is likely that most cases of reported incarceration were that of a romantic partner rather than a mother themselves. Moreover, past research has demonstrated that adverse health outcomes occur for maternal and infant health when a mother herself is incarcerated (Bell et al., 2004; Howard et al., 2011; Sufrin et al., 2019), as well as when incarceration is experienced vicariously through one's partner (Lee et al., 2014; Wildeman et al., 2012; Wildeman et al., 2018). Still, future research should further explore this association using alternative data that specifies who experienced the incarceration to determine whether the association between incarceration and health differs if the event was experienced directly by the mother or vicariously through a partner. Second, the binary indicator of incarceration exposure does not measure potentially important aspects of incarceration (e.g., length of time served, type of facility) that could have implications for the health outcomes evaluated in this study. Third, based on the wording of the survey items we cannot definitively capture whether incarceration and other life stressors temporally preceded the prenatal care and maternal smoking outcomes. Finally, due to the cross-sectional design of the PRAMS survey, we cannot definitively establish the causal order of our key variables of interest. For instance, it is possible that incarceration acts as an initial stressor that in turn exacerbates stressors in other areas of life. While we are not aware of any longitudinal data that measure both incarceration exposure alongside maternal and newborn health, as more data become available in the future, researchers should further investigate the questions posed in this study, as well as determine whether incarceration acts as an initial first push that in turn leads to experiencing other stressful events.

5. Conclusions

The current study extends prior research on the health consequences of incarceration by using individuals who were not incarcerated during pregnancy – but nevertheless experienced a varying number of stressors – as a strategic comparison group. In doing so, we showed that the approach used in prior research may overestimate the unique effects of incarceration on select maternal and newborn health outcomes. At the same time, our results suggest that incarceration is an important and profound stressor that carries important implications for health of both mothers and their newborns. We contend that making strides to reduce the negative ramifications of incarceration requires criminal justice and health practitioners to consider the negative repercussions of incarceration alongside the context of other adverse stressful life events that have occurred in both the short- and long-term and often may occur in conjunction with incarceration. On this point, we recommend that researchers further investigate this issue to gain a deeper understanding of the health consequences of experiencing high rates of life stressors.

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Table 4
Summary of main results.

	Maternal Outcomes (5 coefficients)			Newborn Outcomes (6 coefficients)		
	Null Effects	Positive Effect	Negative Effect	Null Effects	Positive Effect	Negative Effect
Inc vs. No Inc	1/5 (20.0%)	4/5 (80.0%)	0/5 (0%)	1/6 (16.7%)	5/6 (83.3%)	0/6 (0%)
Financial Stress						
Inc vs. No Financial Stress	1/5 (20%)	4/5 (80%)	0/5 (0%)	0/6 (0%)	6/6 (100%)	0/6 (0%)
Inc vs. Low Financial Stress	1/5 (20%)	4/5 (80%)	0/5 (0%)	3/6 (50%)	3/6 (50%)	0/6 (0%)
Inc vs. Medium Financial Stress	3/5 (60%)	2/5 (40%)	0/5 (0%)	3/6 (50%)	3/6 (50%)	0/6 (0%)
Inc vs. High Financial Stress	5/5 (100%)	0/5 (0%)	0/5 (0%)	6/6 (100%)	0/6 (0%)	0/6 (0%)
Spousal Stress						
Inc vs. No Spousal Stress	0/5 (0%)	5/5 (100%)	0/5 (0%)	2/6 (33.3%)	4/6 (66.7%)	0/6 (0%)
Inc vs. Low Spousal Stress	3/5 (60%)	2/5 (40%)	0/5 (0%)	2/6 (33.3%)	4/6 (66.7%)	0/6 (0%)
Inc vs. Medium Spousal Stress	1/5 (20%)	3/5 (60%)	1/5 (20%)	4/6 (66.7%)	2/6 (33.3%)	0/6 (0%)
Inc vs. High Spousal Stress	4/5 (80%)	0/5 (0%)	1/5 (20%)	6/6 (100%)	0/6 (0%)	0/6 (0%)
Emotional Stress						
Inc vs. No Emotional Stress	1/5 (20%)	4/5 (80%)	0/5 (0%)	1/6 (16.7%)	5/6 (83.3%)	0/6 (0%)
Inc vs. Low Emotional Stress	1/5 (20%)	4/5 (80%)	0/5 (0%)	2/6 (33.3%)	4/6 (66.7%)	0/6 (0%)
Inc vs. Medium Emotional Stress	1/5 (20%)	4/5 (80%)	0/5 (0%)	3/6 (50%)	3/6 (50%)	0/6 (0%)
Inc vs. High Emotional Stress	2/5 (40%)	2/5 (40%)	1/5 (20%)	6/6 (100%)	0/6 (0%)	0/6 (0%)

Note: columns may not add up to 100% as a result of rounding.

Abbreviations: Inc = Incarceration.

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Appendix A. Supplementary data

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